

xxxxxxx-5253

For the use of a Registered Medical Practitioner or a Hospital or a Laboratory only

TORLEVA 250 TABLETS

(LEVETIRACETAM TABLETS 250 mg)

BRAND OR PRODUCT NAME

TORLEVA 250 TABLETS

NAME AND STRENGTH OF ACTIVE SUBSTANCE(S)

TORLEVA 250 TABLETS

Each film coated tablet contains:

Levetiracetam.....250mg

Colours: Lake of Indigo Carmine and Titanium dioxide

PRODUCT DESCRIPTION

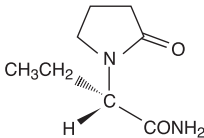
Blue colored, oval shaped, film coated tablets debossed with break line separating '250' and 'MG' on one side and '1014' on other side

DOSAGE FORM

Film coated tablet

PROUCT DESCRIPTION

The chemical name of levetiracetam, a single enantiomer, is (-)-(S)-α-ethyl-2-oxo-1-pyrrolidine acetamide, its molecular formula is C₈H₁₄N₂O₂ and its molecular weight is 170.21. Levetiracetam is chemically unrelated to existing antiepileptic drugs (AEDs). It has the following structural formula:



Levetiracetam is a white to off-white crystalline powder with a faint odor and a bitter taste. It is very soluble in water (104.0 g/100 mL). It is freely soluble in chloroform (65.3 g/100 mL) and in methanol (53.6 g/100 mL), soluble in ethanol (16.5 g/100 mL), sparingly soluble in acetonitrile (5.7 g/100 mL) and practically insoluble in n-hexane

PHARMACODYNAMICS / PHARMACOKINETICS

Pharmacodynamics

Levetiracetam affects intra neuronal Ca2+ levels by partial inhibition of N-type Ca2+ currents and by reducing the release of Ca2+ from intraneuronal stores. In addition, it partially reverses the reductions in GABA- and glycine-gated currents induced by zinc and β-carbolines. Levetiracetam and related analogs had affinity for binding to the synaptic vesicle protein 2A which correlates with the potency of their anti-seizure protection.

Pharmacokinetics

Levetiracetam is readily absorbed from the gastrointestinal tract with a bioavailability of almost 100%; peak plasma concentrations are usually achieved within 1.3 hours of oral doses and steady state achieved after 2 days. Plasma protein binding is minimal at less than 10%. Levetiracetam is not extensively metabolised; about 25% of a dose is metabolised by hydroxylation to inactive metabolites. Around 95% of a dose is excreted as unchanged drug and metabolites in the urine. The plasma elimination half-life has been reported to be about 7 hours in adults and children aged 12 years and over; the half-life may be shorter in younger children. Levetiracetam is distributed into breast milk.

INDICATION

Torleva is indicated as monotherapy in the treatment of partial onset seizures with or without secondary generalization in patients from 16 years of age with newly diagnosed epilepsy.

Torleva is indicated as adjunctive therapy

- in the treatment of partial onset seizures with or without secondary generalization in adults and children from 4 years of age with epilepsy.
- in the treatment of myoclonic seizures in adults and adolescents from 12 years of age with Juvenile Myoclonic Epilepsy.
- in the treatment of primary generalized tonic-clonic seizures in adults and children from 12 years of age with Idiopathic Generalised Epilepsy.

DOSAGE AND ADMINISTRATION

Recommended dose

Levetiracetam therapy can be initiated with either intravenous or oral administration. Conversion to or from oral to intravenous administration can be done directly without titration. The total daily dose and frequency of administration should be maintained.

Film-coated tablets

The film-coated tablets must be taken orally, swallowed with a sufficient quantity of liquid and may be taken with or without food. The daily dose is administered in two equally divided doses.

Adults

- Monotherapy

Adults and adolescents from 16 years of age

The recommended starting dose is 250 mg twice daily which should be increased to an initial therapeutic dose of 500 mg twice daily after two weeks. The dose can be further increased by 250 mg twice daily every two weeks depending upon the clinical response. The maximum dose is 1500 mg twice daily.

- Add-on therapy

Adults (18 years) and adolescents (12 to 17 years) weighing 50 kg or more

The initial therapeutic dose is 500 mg twice daily. This dose can be started on the first day of treatment.

Depending upon the clinical response and tolerability, the daily dose can be increased up to 1500 mg twice daily. Dose changes can be made in 500 mg twice daily increases or decreases every two to four weeks.

Children

The physician should prescribe the most appropriate pharmaceutical form, presentation and strength according to age, weight and dose.

The tablet formulation is not adapted for use in children under the age of 6 years. Levetiracetam oral solution is the preferred formulation for use in this population. In addition, the available dose strengths of the tablets are not appropriate for initial treatment in children weighing less than 25 kg, for patients unable to swallow tablets or for the administration of doses below 250 mg. In all of the above cases levetiracetam oral solution should be used.

Monotherapy

The safety and efficacy of levetiracetam in children and adolescents below 16 years as monotherapy treatment have not been established.

There are no data available.

Add-on therapy for children (4 to 11 years) and adolescents (12 to 17 years) weighing less than 50 kg

Levetiracetam oral solution is the preferred formulation for use in children under the age of 6 years.

The initial therapeutic dose is 10 mg/kg twice daily.

Depending upon the clinical response and tolerability, the dose can be increased up to 30 mg/kg twice daily. Dose changes should not exceed increases or decreases of 10 mg/kg twice daily every two weeks.

The lowest effective dose should be used.

Dose in children 50 kg or greater is the same as in adults.

Dose recommendations for children and adolescents:

Weight	Starting dose 10 mg/kg twice daily	Maximum dose 30 mg/kg twice daily
10 kg ⁽¹⁾	100 mg twice daily	300 mg twice daily
15 kg ⁽¹⁾	150 mg twice daily	450 mg twice daily
20 kg ⁽¹⁾	200 mg twice daily	600 mg twice daily
25 kg	250 mg twice daily	750 mg twice daily
From 50 kg ⁽²⁾	500 mg twice daily	1500 mg twice daily

⁽¹⁾ Children 25 kg or less should preferably start the treatment with levetiracetam 100 mg/ml oral solution

⁽²⁾ Dose in children and adolescents 50 kg or more is the same as in adults

Elderly

Adjustment of the dose is recommended in elderly patients with compromised renal function.

Renal impairment

The daily dose must be individualised according to renal function (*see Section Warnings and Precautions*).

For adult patients, refer to the following table and adjust the dose as indicated. To use this dosing table, an estimate of the patient's creatinine clearance (CLcr) in ml/min is needed. The CLcr in ml/min may be estimated from serum creatinine (mg/dl) determination, for adults and adolescents weighting 50 kg or more, using the following formula:

$$\text{CLcr (ml/min)} = \frac{[140 - \text{age (years)}] \times \text{weight (kg)}}{72 \times \text{serum creatinine (mg/dl)}} \times 0.85 \text{ for women}$$

Then CLcr is adjusted for body surface area (BSA) as follows:

$$\text{CLcr (ml/min/1.73 m}^2\text{)} = \frac{\text{CLcr (ml/min)}}{\text{BSA subject (m}^2\text{)}} \times 1.73$$

Dosing adjustment for adult and adolescent patients weighing more than 50kg with impaired renal function

Group	Creatinine clearance (ml/min/1.73m ²)	Dosage and frequency
Normal	> 80	500 to 1500 mg twice daily
Mild	50-79	500 to 1000 mg twice daily
Moderate	30-49	250 to 750 mg twice daily
Severe	< 30	250 to 500 mg twice daily
End-stage renal disease patients undergoing dialysis ⁽¹⁾	-	500 to 1000 mg once daily ⁽²⁾

⁽¹⁾ A 750 mg loading dose is recommended on the first day of treatment with levetiracetam.

⁽²⁾ Following dialysis, a 250 to 500 mg supplemental dose is recommended.

For children with renal impairment, levetiracetam dose needs to be adjusted based on the renal function as levetiracetam clearance is related to renal function.

This recommendation is based on a study in adult renally impaired patients.

The CLcr in ml/min/1.73 m² may be estimated from serum creatinine (mg/dl) determination using, for young adolescents and children using the following formula (Schwartz formula):

$$\text{CLcr (ml/min/1.73 m}^2\text{)} = \frac{\text{Height (cm)} \times \text{ks}}{\text{Serum Creatinine (mg/dl)}}$$

ks= 0.55 in Children to less than 13 years and in adolescent female; ks= 0.7 in adolescent male

Dosing adjustment for children and adolescent patients weighing less than 50 kg with impaired renal function

Group	Creatinine clearance (ml/min/1.73m ²)	Children and adolescents weighing less than 50 kg
Normal	> 80	10 to 30 mg/kg twice daily
Mild	50-79	10 to 20 mg/kg twice daily
Moderate	30-49	5 to 15 mg/kg twice daily
Severe	< 30	5 to 10 mg/kg twice daily
End-stage renal disease patients undergoing dialysis	-	10 to 20 mg/kg once daily ⁽²⁾⁽³⁾

⁽¹⁾ Levetiracetam oral solution should be used for doses under 250 mg and for patients unable to swallow tablets.

⁽²⁾ A 15 mg/kg (0.15 ml/kg) loading dose is recommended on the first day of treatment with levetiracetam.

⁽³⁾ Following dialysis, a 5 to 10 mg/kg (0.05 to 0.10 ml/kg) supplemental dose is recommended.

Hepatic impairment

No dose adjustment is needed in patients with mild to moderate hepatic impairment. In patients with severe hepatic impairment, the creatinine clearance may underestimate the renal insufficiency. Therefore, a 50% reduction of the daily maintenance dose is recommended when the creatinine clearance is < 60 ml/min/1.73m² (*see Section Warnings and Precautions*).

Torleva tablets are available in three strengths of 250mg, 500mg and 1000mg. They are not able to deliver all approved dose regimens of Levetiracetam; other approved dosage forms and strengths of Levetiracetam should be used in such cases.

CONTRAINDICATIONS

Hypersensitivity to Levettiracetam or other pyrrolidone derivatives or any of the excipients.

WARNINGS AND PRECAUTIONS

Potential for an increase in risk of suicidal thoughts or behaviors.

In accordance with current clinical practice, if TORLEVA has to be discontinued it is recommended to withdraw it gradually.

Due to its complete and linear absorption, plasma levels can be predicted from the oral dose of Levetiracetam expressed as mg/kg bodyweight. There is no need therefore for plasma level monitoring of Levetiracetam.

To date, there are no data to support the use of Levetiracetam in patients less than 4 years of age due to insufficient data on safety and efficacy.

No data on the interaction of Levetiracetam with alcohol are available.

Impaired renal function

The administration of TORLEVA to patients with renal impairment may require dose adaptation. Monitoring of renal function in severe hepatic impaired patients is recommended before dose selection (refer DOSAGE AND ADMINISTRATION).

Acute Kidney Injury

The use of levetiracetam has been rarely associated with acute kidney injury, with a time to onset ranging from a few days to several months.

Carcinogenicity/Mutagenicity

There was no evidence of carcinogenicity following dietary administration of Levetiracetam to rats at doses up to 1800mg/kg/day for 104 weeks and to mice at doses up to 960mg/kg/day for 80 weeks. In rats, the highest dose achieved systemic exposure (plasma AUC) approximately 6 times that in humans at the maximal recommended clinical dose of 3000mg/day. In mice, the highest dose achieved systemic exposure (plasma AUC) approximately twice that in humans at the maximal recommended clinical dose. Because adequate doses have not been studied in mice, the carcinogenic potential in this species has not been fully evaluated.

Assays for gene mutations (reverse mutation in bacteria, Chinese hamster ovary/HGPRT locus assay) and chromosomal damage (Chinese hamster ovary cells in vitro, mouse micronucleus test) did not provide evidence of genotoxic potential of Levetiracetam. The hydrolysis product and major human metabolite (ucb L057) was not mutagenic in bacterial reverse mutation assays or the in vitro mouse lymphoma assay.

Impairment of Fertility

There are no human data on the effects of TORLEVA on male or female fertility. No adverse effects on male or female fertility or reproductive performance were observed in rats at oral doses of Levetiracetam up to 1800mg/kg/day (corresponding to approximately 6 times the maximal recommended clinical dose on a mg/m2 basis) administered for at least two weeks prior to, and throughout, mating.

Effects on ability to drive or operate machinery

No studies on the effects on the ability to drive and use machines have been performed. Due to possible different individual sensitivity, some patients might experience, at the beginning of treatment or following a dosage increase, somnolence or other CNS related symptoms. Therefore, caution is recommended in those patients when performing skilled tasks, e.g. driving vehicles, or operating machinery.

INTERACTIONS WITH OTHER MEDICAMENTS

Antiepileptic medicinal products

Levetiracetam did not influence the serum concentrations of existing antiepileptic medicinal products (phenytoin, carbamazepine, valproic acid, phenobarbital, lamotrigine, gabapentin and primidone) and that these antiepileptic medicinal products did not influence the pharmacokinetics of Levetiracetam.

Levetiracetam clearance was higher in children taking enzyme-inducing antiepileptic medicinal products. However, dose adjustment is not required.

Probenecid

Probenecid increased the plasma levels of an inactive metabolite of Levetiracetam. The clinical relevance of elevated levels of it is not known, therefore it has suggested that caution is warranted.

Oral contraceptives and other pharmacokinetics interactions

Levetiracetam 1,000 mg daily did not influence the pharmacokinetics of oral contraceptives (ethinylestradiol and levonorgestrel); endocrine parameters (luteinizing hormone and progesterone) were not modified. Levetiracetam 2,000 mg daily did not influence the pharmacokinetics of digoxin and warfarin; prothrombin times were not modified. Co-administration with digoxin, oral contraceptives and warfarin did not influence the pharmacokinetics of Levetiracetam.

Food

The extent of absorption of Levetiracetam was not altered by food, but the rate of absorption was slightly reduced.

STATEMENT ON USAGE DURING PREGNANCY AND LACTATION

Pregnancy

Levetiracetam is not recommended during pregnancy and in women of childbearing potential not using contraception unless clearly necessary.

As with other antiepileptic medicinal products, physiological changes during pregnancy may affect Levetiracetam concentration. Plasma concentrations of

Levetiracetam decreased during pregnancy.

Appropriate clinical management of pregnant women treated with Levetiracetam should be ensured.

Discontinuation of antiepileptic treatments may result in exacerbation of the disease which could be harmful to the mother and the foetus.

Breastfeeding

Levetiracetam is excreted in human breast milk. Therefore, breast-feeding is not recommended.

ADVERSE EFFECTS / UNDESIRABLE EFFECTS

Incidence (%) of treatment-emergent adverse events in adults experiencing partial onset seizures by body system (adverse events occurred in at least 1% of Levetiracetam-treated patients):

Body as a Whole

Asthenia, headache, infection and pain

Digestive System

Anorexia

Nervous System

Somnolence, dizziness, depression, nervousness, ataxia, vertigo, amnesia, anxiety, hostility, paresthesia and emotional lability

Respiratory System

Pharyngitis, rhinitis, cough increased and sinusitis

Special Senses

Diplopia

Renal and urinary disorders:

Frequency rare: acute kidney injury

Musculoskeletal and connective tissue disorder:

Frequency rare: Rhabdomyolysis and blood creatine phosphokinase increased. *

*Prevalence is significantly higher in Japanese patients when compared to non-Japanese patients.

Cases of Encephalopathy have been rarely observed after levetiracetam administration. These undesirable effects generally occurred at the beginning of the treatment (few days to a few months) and were reversible after treatment discontinuation.

Incidence (%) of treatment-emergent adverse events in pediatric patients ages 4-16 years experiencing partial onset seizures by body system (adverse events occurred in at least 2% of Levetiracetam-treated patients):

Body as a Whole

Accidental injury, asthenia, pain, flu syndrome, face edema, neck pain and viral infection

Digestive System

Vomiting, anorexia, diarrhoea, gastroenteritis and constipation

Haemic and Lymph System

Echymosis

Metabolic / Nutr Dis

Dehydration

Nervous system

Somnolence, hostility, nervousness, personality disorder, dizziness, emotional lability, agitation, depression, vertigo, reflexes increased and confusion

Respiratory System

Rhinitis, cough increased, pharyngitis and asthma

Skin and Appendages

Pruritus, skin discoloration and vesiculobullous rash

Special Senses

Conjunctivitis, amblyopia and ear pain

Urogenital System

Albuminuria and urine abnormality

OVERDOSE AND TREATMENT

Symptoms

Somnolence, agitation, aggression, depressed level of consciousness, respiratory depression and coma were observed with Levetiracetam overdoses.

Management of overdose

After an acute overdose, the stomach may be emptied by gastric lavage or by induction of emesis. There is no specific antidote for levetiracetam. Treatment of an overdose will be symptomatic and may include haemodialysis. The dialyser extraction efficiency is 60 % for levetiracetam and 74 % for the primary metabolite

STORAGE CONDITIONS

Store below 30°C.

PACKAGING AVAILABLE

TORLEVA 250 TABLETS are packed in PVC/PVDC pack of 10 tablets. Such packs containing 10 Tablets are packed into boxes of 60's and 100's. Not all presentations may be available locally.

DATE OF REVISION OF PACKAGE INSERT

24th July 2025



Manufactured by:
TORRENT PHARMACEUTICALS LTD
Indrad - 382 721, Dist. Mehsana, INDIA.

PRODUCT NAME	:	Torleva 250 Tablets	COUNTRY : Malaysia	LOCATION : Indrad	Supersedes A/W No.:			
ITEM / PACK	:	Insert	NO. OF COLORS: 1	REMARK :				
DESIGN STYLE	:		PANTONE SHADE NOS.:	SUBSTRATE :				
CODE	:	xxxxxxxx-5253	 Black	Activities	Department	Name	Signature	Date
DIMENSIONS (MM) (LxWxH)	:	180 x 240		Prepared By	Pkg.Dev			
ART WORK SIZE	:	S/S		Reviewed By	Pkg.Dev			
DATE	:	28-07-2025		Approved By	Quality			